BOLT UBC

# FIRST BYTE 2026

## CASE PACKAGE



BYTE

BOLT UBC MASCOT

# BOLT UBC FIRST BYTE 2026

## Disclaimer

BOLT UBC

It is the competitors' responsibility to review and uphold applicable standards of fair competition.

Instances of misconduct, such as cheating, plagiarism, resubmitting the same code/slides, or impersonating a candidate, will result in the competitor and their team being automatically disqualified.

If you have any questions, please reach out to us at:
developer@boltubc.com

# BOLT UBC FIRST BYTE 2026

## Table of Contents

BOLT UBC


**Disclaimer** 2

**Table of Contents** 3

**Competition Schedule** 4

**Case**

- Introduction 5
- Company Background 6
- Industry Overview 7
- Company Situation 8
- Revenue & Customers 9
- Operating Costs 10
- Airports & Gates 11
- The Challenge 14
- Key Ask 15
- Data Description 16
- Guiding Questions 17

**Case Deliverables**

- Guidelines and Deadline 18
- Judging Criteria 19
- Rounds 20

# BOLT UBC FIRST BYTE 2026

## Competition Schedule

**October 7th, Wednesday**

12:00 PM | Case Release

**October 13th, Tuesday**

11:59 PM | Workshop for Beginners

**October 15th, Thursday**

11:59 PM | Case Submission Deadline

**October 16th, Friday**

By 6:00 PM | Semi-Finalist Announcement

**Final Presentation - October 17th, Saturday (In-person)**

**Location: Sauder School of Business**

10:30 - 11:00 AM Sign-in Opens

11:00 - 11:30 AM Opening Remarks and Room Assignment

11:30 - 1:00 PM Preliminary Round

1:00 - 2:00 PM Lunch + Open Networking

2:00 - 3:00 PM Finalist Presentations

3:00 - 3:30 PM Prizes + Closing Remarks

## Introduction

In a country as large as Canada, air travel is more than a convenience. It connects families across provinces, moves business travellers between economic centres, and keeps smaller and northern communities linked to the rest of the country. Yet behind every boarding pass is one of the thinnest-margin businesses in the world: airlines spend heavily on fuel, crews, aircraft, and airport space long before a single passenger boards.

NorthByte Airlines finds itself at a turning point. After more than a decade of steady growth, the Vancouver-based carrier just recorded its highest revenue ever, and yet it is now losing tens of millions of dollars a year. Bookings are up, customers are loyal, and new destinations keep being added to the map.

Leadership has plenty of theories but no clear answer. With the next budget cycle approaching, NorthByte Airlines must figure out what is really driving its losses before deciding where to invest and where to pull back.

## Company Background

NorthByte Airlines is a fully domestic carrier serving 20 airports across Canada, from Victoria to St. John's and as far north as Whitehorse and Yellowknife. Most flights connect through four major airports (Vancouver, Calgary, Toronto, and Montreal), while regional routes link smaller cities into the network.

### Mission

To connect every corner of Canada with safe, affordable, and friendly air travel, bringing people closer to the places and people that matter to them.

### Values

- **Safety First:** Every decision, from scheduling to maintenance, starts with the safety of our passengers and crew.
- **Connecting Communities:** Smaller cities deserve reliable air service too. NorthByte takes pride in serving communities that larger airlines overlook.
- **Everyday Value:** Fair fares, transparent fees, and a travel experience that respects our customers' time and money.
- **Teamwork & Respect:** Our pilots, cabin crew, ground staff, and airport partners succeed together.

## Industry Overview

Canada's domestic air travel market has fully recovered from the pandemic, with passenger numbers once again reaching record highs. The market is dominated by two large national carriers, while a rotating group of smaller and low-cost airlines competes for price-sensitive travellers. Over the past decade, several newer carriers have entered the market with ambitious expansion plans, only to scale back or shut down within a few years.

Running an airline in Canada comes with unique challenges. Distances between cities are long, and many communities are small, making it difficult to fill aircraft on every route. Harsh winters bring de-icing, delays, and cancellations, particularly in Atlantic Canada and the North. Airport fees in Canada are among the highest in the world, and airlines must lease gates at each airport they serve, whether or not those gates are busy.

On top of this, airlines face rising costs across the board. Jet fuel prices remain volatile, a national pilot shortage has pushed up crew wages, and passenger protection rules require airlines to compensate or refund travellers when flights are significantly delayed or cancelled. In this environment, it is crucial for airlines to know where they make money, and where they do not.

## Company Situation

NorthByte's fiscal year runs from September 1 to August 31. On paper, the fiscal year that just ended, FY2025-26, should have been NorthByte's best yet. Total revenue grew by about 5% compared to FY2024-25, driven by more passengers and the biggest expansion in NorthByte's recent history. In spring 2025, NorthByte launched three new connections and grew its schedule to carry more travellers than ever. The expansion was celebrated internally as a major milestone, adding new cities (London and Moncton) to NorthByte's map for the first time in five years.

However, the growth did not reach the bottom line. Over the same period, NorthByte's costs grew faster than its revenue. NorthByte's flights were profitable through the fall of FY2024-25, earning a margin of roughly 1% to 10% each month, but they have lost money every month since February 2025. In FY2025-26, NorthByte lost about $40 million on its flights, roughly four times its loss the year before. That figure does not yet include what NorthByte pays to lease its airport gates.

NorthByte's leadership knows that something needs to change, but company-wide reports have not made it clear where the money is being lost. Fuel and labour costs have both risen, winter disruptions have increased refunds, and competitors continue to invest in their loyalty programs. Before committing its limited budget, NorthByte wants an independent, data-driven view of what is really behind its losses.

## Revenue and Customers

Ticket sales make up the vast majority of NorthByte's revenue. Fares vary by route, distance, and season, with demand peaking during the summer months and the December holidays. Passengers travelling between two cities without a direct NorthByte flight can connect through one of NorthByte's hubs; in that case, a single booking includes two or more flights, and each flight is recorded as its own ticket.

Customers book through NorthByte's website, its mobile app, corporate travel portals, or online travel agencies, and can pay by credit card, debit, travel voucher, or a combination of NorthByte Rewards points and cash.

On top of the base fare, NorthByte earns ancillary revenue from optional extras:

- Extra Baggage: fees for checked bags beyond the fare's allowance.
- Seat Upgrade: payments for extra legroom or preferred seats.
- Change Fee: charged when a passenger changes their travel date or time.
- Food & Onboard Purchases: snacks, meals, and drinks bought during the flight.

Ancillary revenue has become increasingly important across the airline industry, and NorthByte's leadership views it as a key area for growth.

Not all revenue stays with NorthByte. Under its customer promise, NorthByte refunds a passenger's full fare when their flight is cancelled or delayed by more than three hours. Refunds are recorded as separate transactions against the original booking. Some parts of NorthByte's network are far more exposed to weather disruptions than others, especially during the winter.

Finally, many of NorthByte's customers belong to NorthByte Rewards, its free membership program. Members are placed into Blue, Silver, Gold, or Platinum tiers based on how often they fly, with higher tiers receiving perks such as priority boarding, lounge access, and earning points per flight. Travelers who have not joined the program are recorded as Non-Members.

## Operating Costs

Every flight NorthByte operates carries a cost, whether it departs full or nearly empty. NorthByte's finance team records four categories of operating cost for every flight it operates:

- **Fuel:** jet fuel burned on each flight. Longer routes burn significantly more fuel, and prices fluctuate with global oil markets. NorthByte's average fuel cost per flight rose by about 3% in FY2025–26 compared to FY2024–25.
- **Crew:** wages and expenses for pilots and cabin crew. Following the new pilot agreement, NorthByte's average crew cost per flight rose by about 4% in FY2025–26.
- **Maintenance:** scheduled inspections, repairs, and parts. Maintenance needs increase with the number of flights and are heavier in winter.
- **Food:** catering loaded onto the aircraft, including complimentary snacks and drinks and items stocked for onboard sale.

Because most of these costs depend on the number of flights rather than the number of passengers, a flight with half its seats empty costs nearly as much to operate as a full one. NorthByte's finance team has historically reviewed costs at the company level, by category, rather than route by route.

## Airports & Gates

NorthByte serves 20 airports across Canada. Four of them (Vancouver, Calgary, Toronto, and Montreal) act as hubs, with many routes and connecting passengers. The remaining airports are regional stations served by one or a few routes.

| City | Airport Code | Province/Territory | Role |
|---|---|---|---|
| Vancouver | YVR | British Columbia | Hub |
| Calgary | YYC | Alberta | Hub |
| Toronto | YYZ | Ontario | Hub |
| Montreal | YUL | Quebec | Hub |
| Victoria | YYJ | British Columbia | Regional |
| Kelowna | YLW | British Columbia | Regional |
| Abbotsford | YXX | British Columbia | Regional |
| Prince George | YXS | British Columbia | Regional |
| Edmonton | YEG | Alberta | Regional |

## Airports & Gates Continued

| | | | |
|---|---|---|---|
| **Saskatoon** | YXE | **Saskatchewan** | **Regional** |
| **Regina** | YQR | **Saskatchewan** | **Regional** |
| **Winnipeg** | YWG | **Manitoba** | **Regional** |
| Ottawa | YOW | **Ontario** | **Regional** |
| London | YXU | Ontario | Regional |
| Quebec City | YQB | Quebec | Regional |
| Halifax | YHZ | Nova Scotia | Regional |
| Moncton | YQM | New Brunswick | Regional |
| St. John's | YYT | Newfoundland and Labrador | Regional |
| Whitehorse | YXY | Yukon | Regional |
| Yellowknife | YZF | Northwest Territories | Regional |

## Airports & Gates Continued

To operate at an airport, NorthByte leases departure gates on an annual basis. Each lease has a fixed yearly cost that NorthByte pays regardless of how many flights use the gate, and a single gate can be shared by several routes departing from that airport. NorthByte only pays for gates at the airport a flight departs from; arriving flights add no gate cost. Lease prices vary widely: gates at major hubs cost far more than gates at smaller regional airports, and prices are renegotiated every year.

Unlike the rest of NorthByte's finances, gate leases do not follow NorthByte's fiscal year. Each lease runs from January 1 of one year to January 1 of the next. For example, the "2024-2025" lease covers January 1, 2024, to January 1, 2025, and the "2025-2026" lease covers January 1, 2025, to January 1, 2026. NorthByte's records for its 2026-2027 leases were lost in an accounting error. For this analysis, assume each airport's total gate cost is split evenly across all NorthByte flights departing from that airport.

Gate usage is uneven across the network. At some airports, station managers report that every gate NorthByte leases is in near-constant use. At others, leased gates sit empty for much of the day.

## The Challenge

NorthByte Airlines is growing, but it is not growing profitably. Revenue is at a record high, yet NorthByte now spends more to run its flights than it earns from them. Without a clear picture of where its money is being lost, NorthByte risks spending its limited budget on the wrong fix.

The challenge is to look past company-wide totals and understand where NorthByte actually makes and loses money. Many factors play a role, but not every factor matters equally. Any solution must also stay true to NorthByte's values: protecting safety, treating customers fairly, and continuing to connect communities across Canada, while returning the airline to a sustainable level of profit.

## Case

BOLT UBC

### Key Ask

Your team has been hired as external data consultants to help NorthByte Airlines identify what is driving its losses and recommend how to restore profitability. Along with the information already provided, NorthByte has given your team three datasets covering two full fiscal years of operations (September 2024 to August 2026): customer transactions, operating costs, and airport gate leases.

NorthByte's leadership team has stated that there are some inconsistencies within the data, as the datasets were exported from three separate systems. One of these is that route codes are not recorded in the same format across, or even within, the datasets. Keep in mind that gate leases run from January 1 to January 1 of the next year rather than on NorthByte's fiscal year, and that the 2026-2027 gate lease records are missing. Use your best judgment when estimating these costs, and state your assumptions clearly.

Your role is to clean this data, analyze trends, uncover insights, and recommend supported, actionable solutions for NorthByte's leadership team.

## Data Description

This case package includes three datasets. Please refer to the accompanying Data Dictionary for a full description of each variable.

- **transactions.csv** contains every customer transaction recorded between September 2024 and August 2026, including ticket sales, ancillary purchases, change fees, and refunds. Each row is one transaction linked to a booking, a route, the customer's NorthByte Rewards tier, the booking channel, and the payment method. Each row is one transaction, identified by a transaction ID.

- **costs.csv** contains NorthByte's operating costs for every flight over the same period, broken down into fuel, crew, maintenance, and food, along with the departure and arrival city of each route. Each flight appears once for each cost category.

- **gates.csv** lists every departure gate NorthByte leased at each airport for the 2024-2025 and 2025-2026 lease years, the routes assigned to each gate, and the annual cost of each lease. Each lease year runs from January 1 to January 1 (for example, 2024-2025 covers January 1, 2024 to January 1, 2025), and gate costs apply only to the departure airport and are split evenly across all flights departing from it. Records for the 2026-2027 lease year were lost in an accounting error and are not included.

## Guiding Questions

1. How did NorthByte's revenue and costs change between FY2024-25 and FY2025-26? When did NorthByte's results start to decline, and which changes explain most of the drop?

2. How does financial performance differ across NorthByte's network? Are some parts of the business carrying others?

3. How much do rising fuel and crew costs, refunds, and the NorthByte Rewards program contribute to the decline? Which factors matter most?

4. What role do ancillary revenue, refunds, and connecting passengers play in how NorthByte earns its money?

5. How did NorthByte's gate lease costs change from the 2024-2025 lease year to the 2025-2026 lease year? Based on these trends, what would you expect NorthByte to pay for its gates in 2026-2027?

6. What short-term and long-term actions would you recommend, and how would they affect NorthByte's customers, employees, and the communities it serves?

# BOLT UBC FIRST BYTE 2026

## Case Deliverables

BOLT UBC

### Guidelines and Deadline

A slide deck of analysis of the issues, possible alternatives, and final recommendation(s) that will accompany your presentation if you make it past qualifiers. Each team will have **10 minutes to present** and 5 minutes of Q&A.

- Naming convention: [TeamName_FirstByte2026].pdf
  - Example: TeamBOLT_FirstByte2026.pdf
- File Format: PDF ONLY

Please submit your deck as a pdf file via this ***form*** by **October 15th, Thursday, 11:59 PM** (midnight)

ONLY ONE team member has to upload the submission.

Note that **NO changes are allowed after the submission deadline**. The same slide deck will be used on Presentation Day.
No late submissions will be accepted.

## Case Deliverables

BOLT UBC

### Judging Criteria

| | |
|---|---|
| **Data Analysis** | **35%** |
| **Business Recommendation** | **35%** |
| **Presentation** | **30%** |

We recommend taking screenshots of your data analytics takeaways and embedding those insights into your presentation.

# BOLT UBC FIRST BYTE 2026

## Case Deliverables

BOLT UBC

### First Round

Case submissions will be evaluated and the Semi-Finalist teams will be announced. Your team will be notified via Email **by 6:00 PM on October 16th, Friday** if you have passed this round.

Semi-Finalist teams will be invited to present their case submission live in Henry Angus on Saturday October 17th, in front of a panel of judges.

---

### Semi-Finalist Round

All Semi-Finalist teams will be divided into different judging breakout rooms randomly. Each team will present in their respective breakout rooms.

**Due to the tight schedule, late teams will be disqualified.**

Each team will have 10 minutes to present and 5 minutes of Q&A.

The breakout room judges will select the top team in their breakout room to advance to the Final Round.

---

### Final Round

Finalist teams will present to a panel of judges. Each team will have 10 minutes to present and 5 minutes of Q&A.